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May 17, 1971
SUPERSEDING
Int. Fed. Spec. DDD-R-00271A (GSA-FSS)
November 24, 1953 and
INTERIM REVISION OF
Fed. Spec. DDD-R-271A
January 28, 1948

INTERIM FEDERAL SPECIFICATION

RIBBONS; COMPUTING AND RECORDING MACHINES

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendation for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification DDD-R-271A, dated January 28, 1948.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers record ribbons of cotton and nylon fabric for use on computing and recording machines. For the purpose of this specification, computing and recording machines include: adding machines, adding and subtracting machines, printing calculating machines, machines designed to maintain bookkeeping records, cash registers, check writing machines, shorthand writing machines, and miscellaneous recording instruments.

1.2 Classification. Computing and recording machine ribbons shall be of the following compositions, as specified (see 6.2):

Composition A - Cotton
Composition B - Nylon

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

UU-P-121 - Paper, Bond and Writing, White and Colored.
PPP-B-566 - Boxes; Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Set-Up.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 191/GEN(General Provisions) - Textile Test Methods.
Fed. Test Method 191/2811 - pH of Textiles Electrometric.
Fed. Test Method 191/5010 - Determination of Length of Textile Materials.
Fed. Test Method 191/5020 - Determination of Width of Textile Materials.
Fed. Test Method 191/5030 - Determination of Thickness of Textile Materials.
Fed. Test Method 191/5050 - Yarns per Inch in Woven Cloth.
Fed. Test Method 191/5100 - Breaking Strength and Elongation, of Woven Cloth; Grab Method.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Agent, 1616 P Street, N.W., Washington, D. C. 20036.

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Weave. The fabric used in the ribbons shall have a plain weave.

3.2 Edges. Edges of ribbons shall be straight and parallel to the warp threads. Edges of cotton ribbons shall be woven or treated to prevent fraying, and edges of nylon ribbons shall be fused.

3.3 Yarns per inch.

3.3.1 Composition A - Cotton. There shall be not less than 132 yarns per inch in both the warp and filling direction; total yarns shall be not more than 276 (yarns over the minimum of 264 may be distributed in either the warp or filling direction). (See 4.1.1.2.)

3.3.2 Composition B - Nylon. There shall be not less than 164 yarns per inch in the warp direction and 106 yarns per inch in the filling direction; total yarns shall be not more than 282 (yarns over the minimum of 270 may be distributed in either the warp or filling direction). (See 4.1.1.2.)

3.4 Thickness and breaking strength.

3.4.1 Thickness. Unless otherwise specified, the cotton fabric for composition A ribbons shall be not more than 0.0058 inch thick, and the nylon fabric for composition B ribbons shall be not more than 0.0052 inch thick. (See 4.1.1.2.)

3.4.2 Breaking strength. The breaking strength of the cotton fabric shall be not less than 50 pounds in both the warp and filling direction, and the breaking strength of the nylon fabric shall be not less than 100 pounds in the warp direction and 90 pounds in the filling direction. (See 4.1.1.2.)

3.5 pH. The pH of the uninked fabric used in the ribbons shall be not less than 5.0, nor more than 7.5, when tested as described in 4.1.1.2.

3.6 Inking and color. Ribbons shall be evenly inked. Impressions made by ribbons shall be single-color or two-color, as specified. Unless otherwise specified, two-color ribbons shall have an equal width for each color. (See 6.2.)

3.6.1 Degree of inking. Unless otherwise specified (see 6.2), cotton ribbons shall be furnished in heavy inking, and nylon ribbons shall be furnished in medium-heavy inking.

3.7 Length and width.

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3.7.1 Length. The nominal ribbon length shall be as specified (see 6.2), and shall be not less than an amount equal to one-half inch for each one yard of ribbon. Ribbons shall be furnished in continuous lengths, without joining or splicing. (See 4.4.1.1.)

3.7.2 Width. The ribbon width shall be uniform and as specified (see 6.2). A plus tolerance of 1/64 inch is permitted. (See 4.4.1.2.)

3.8 Performance.

3.8.1 Wear down. When the wear down test is run as specified in 4.4.2.4, the first line shall be clear, clean, and legible to persons of normal or corrected 20/20 vision under an illumination of 25 foot-candles. The number of lines that are of good intensity, distinct, and easily legible shall be as shown in table I for the respective ribbon. Both portions of the two-colored ribbon shall be subjected to the wear down test.

3.8.2 Recovery. After making the wear down test, the ribbon shall be tested for recovery as described in 4.4.2.2. The recovery line shall equal the intensity of the designated comparison line of the wear down test as shown in table I for the respective ribbon. Both portions of the two-colored ribbon shall be subjected to the recovery test.

TABLE I. Wear down and recovery requirements

Ribbon	WEAR DOWN (total number of lines required)	RECOVERY (comparison line of the wear down test)
Composition A Cotton		
Single-colored	120	100
Two-colored	70	60
Composition B nylon		
Single-colored	170	140
Two-colored	100	85

3.8.3 Type filling. There shall be no evidence of broadening or feathering of type impressions when tested as described in 4.4.2.3. For the purpose of this specification, type filling is defined as the filling of the type face with fibers or ink from the ribbon, resulting in a gradual broadening or feathering of typed impressions.

3.8.4 Oiliness. There shall be no evidence of oil staining when the ribbon is examined as described in 4.4.2.4.

3.9 Spools, reversing attachments, and accessories. Each ribbon shall be designed to operate satisfactorily with the ribbon feed and reversing mechanisms of the machine specified (see 6.2). This requirement includes applicable spool(s) or core(s), reversing attachments, hooks, sewed loops, tapered ends, and other accessories necessary for proper operation.

3.10 Shelf life. Ribbons in original unopened containers and not subjected to extremes of temperature and humidity, shall retain their serviceability as defined by this specification for not less than 24 months from date of manufacture.

3.11 Workmanship. Ribbons shall be evenly woven and free from frayed edges, slits, slubs, knots, tears, loops, cuts, holes, or other imperfections which might affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Examination and testing of basic fabric material. The supplier is responsible for performing the following tests on the basic fabric (uninked)used in manufacturing the ribbons.

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4.1.1.1 Sampling of fabric for tests. The lot shall consist of one shipment of fabric of the same description, manufactured by the same source under essentially the same condition, and identified as used in fabricating ribbons in accordance with this specification. The lot shall be expressed in square yards of fabric. The sample unit shall be one square yard of fabric, and shall be taken from each lot in accordance with table II.

TABLE II. Fabric sample size

Square yards of fabric in the lot	Sample size
100 or less	2
101 to 300	3
301 and over	4

4.1.1.2 Testing of fabric. Samples taken in accordance with 4.1.1.1 shall be tested in accordance with the following test methods described in Fed. Test Method Std. No. 191.

Test	Method	Reference paragraph
Yarns per inch	5050	3.3
Thickness	5030 (See Note)	3.4
Breaking strength	5100	3.4
pH	2811	3.5

Note: Exceptions to method 5030 shall be as follows: The dial of the gage shall be graduated to read directly to 0.0001 inch. The diameter of the gage pressure foot shall be 0.407 ± 0.001 inch. The gage pressure foot shall not rest on edges of fabric. The average thickness shall be reported to the nearest 0.0001 inch.

4.2 End-item sampling.

4.2.1 Lot. For purpose of end-item sampling, a lot shall consist of all ribbons presented for inspection at one time as defined in MIL-STD-105.

4.2.2 Sampling of end-item for visual defects. A random sample of ribbons shall be taken from each lot in accordance with MIL-STD-105, inspection level S-2. The acceptable quality level (AQL) shall not exceed 6.5 percent defective. The sample unit shall be one ribbon.

4.2.3 Sampling of end-item for preparation for delivery requirements. For examination of interior packages the sample unit shall be one shipping container fully prepared for delivery, and selected at random just prior to the closing operation. Sampling shall be in accordance with MIL-STD-105. Closure defects shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end-item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

4.2.4 Sampling of end-item tests. A random sample of ribbons shall be taken from each lot in accordance with MIL-STD-105, inspection level S-1. The AQL shall not exceed 4.0 percent defective. The sample unit shall be one ribbon.

4.3 Examination.

4.3.1 Visual examination. Samples taken in accordance with 4.2.2 shall be examined for defects listed in table III.

TABLE III. Classification of defects

Examine	Defects
General	Weave not plain. (3.1) Edges not straight, or parallel to the warp. (3.2) Edges not woven or treated to prevent fraying. (3.2) Color not as specified. (3.6) Ribbon spliced or joined. (3.7.1) Ribbon width not uniform. (3.7.2)
Workmanship	Imperfections which might affect serviceability. (3.11)

4.3.2 Examination of preparation for delivery. Samples taken in accordance with 4.2.3 shall be examined to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table IV.

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TABLE IV. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, or distortion of containers.

4.4 End-item tests. Samples taken in accordance with 4.2.4 shall be subjected to the following tests. Unless otherwise specified, tests shall be conducted in an atmosphere maintained at a temperature of 23°C. ± 1.1°C. (73.5°F. ± 2°F.) and a relative humidity of 50 ± 4 percent. Test supplies and equipment, and the sample ribbons shall be conditioned in these atmospheric conditions for a minimum of 12 hours prior to tests.

4.4.1 Dimensional tests.

4.4.1.1 Length. Length shall be determined in accordance with method 5010 of Fed. Test Method Std. No. 191 (see 3.7.1).

4.4.1.2 Width. Width shall be determined in accordance with method 5020 of Fed. Test Method Std. No. 191 except the rule shall be graduated to 1/64 inch (see 3.7.2).

4.4.2 Performance tests. The performance tests shall be made on an electric typewriter, IBM Model C (equipped for fabric ribbon), or equivalent, with pica type, code 001, type mark P, or equal. The typewriter shall have a platen with a diameter of 1.74 ± 0.01 inches and a hardness value of 92 ± 2, as determined with a Shore durometer, Model A-2, or equal. An average of five readings taken at widely separated points shall be used for the hardness value. The typewriter platen ratchet shall have a *vertical line spacing of six lines per inch. The Multiple Copy Control shall be set* at the smallest opening and the Ribbon Position Lever shall be at stencil position.

The impression force of the capital letter "B" shall be adjusted through the use of aluminum foil having a thickness of 0.0037 ± 0.0003 inch. Thickness shall be determined with a dead-weight micrometer having two plane faces, the smallest of which is circular and 0.25 to 0.33 square inch in area. An average of five readings taken at widely separated points shall be used for the thickness dimension. The impression force of the "B" shall be adjusted to provide an increase in the foil's embossed thickness of 0.0011 to 0.0013 inch.

Emboss five blocks consisting of five single-spaced lines of ten "B's" on the previously measured aluminum foil at widely separated points. With the dead-weight micrometer, measure the thickness of each embossed block of images. The average of the five readings shall be used for the foil's embossed thickness.

4.4.2.1 Wear down. Unwind about a yard of the ribbon and lay the spool at one side of the machine, with the end of the ribbon across the machine. If the ribbon is too wide for the typewriter ribbon carrier, it shall be cut down to a width no greater than 9/16 inch. This must be done with care so as to remove as little ink as possible, and may be limited to a length sufficient to perform the wear down, recovery, and type filling tests. Two-colored ribbons shall be cut with an equal width of each color.

By some suitable means, fasten the ribbon to opposite ends of the carriage, with the ribbon in contact with the paper under just enough tension (approximately 50 grams) to permit writing. The ribbon shall be adjusted at such a height that the type will strike the portion of the ribbon normally used in typing. In the case of two-color ribbons the type shall strike the middle of the colored part under test. Insert a sheet of white bond paper (32 pounds, 25 percent rag, Fed. Spec. UU-P-121) which is at least 21 inches long (if necessary, sheets of letter-sized paper may be used) and then write a line of 30 capital B's. Push carriage, bearing the ribbon with it, back to the starting point, shift the paper a single-space line and again write the same letter over the identical part of the ribbon. Repeat until the required number of lines for the type ribbon (see Table I) under test have been written. It is necessary that the whole number of lines be typed without interruption. Inspect for compliance with 3.8.1.

4.4.2.2 Recovery. Leave the ribbon in the machine for 20 minutes, then type another line of the letter B over the identical part of the ribbon. Examine the intensity of this line for compliance with 3.8.2.

4.4.2.3 Type filling. Unfasten the ribbon from the carriage. With as little handling as possible, install the ribbon on the typewriter in the usual manner. After thoroughly cleaning the capital letter B, make 300 impressions with it under the normal travel of the ribbon. Examine for compliance with 3.8.3.

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4.4.2.4 Oiliness. After 24 hours the test sheet (4.4.2.1 through 4.4.2.3) shall be examined for evidence of oil stains around the letters and on the reverse side of the sheet (see 3.8.4).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each ribbon shall be wrapped in waxed paper, polyethylene, cellophane or other suitable material and enclosed in an individual metal, paperboard, fiberboard or other suitable container.

5.1.1.2 Intermediate packaging. Unless otherwise specified (see 6.2), twelve unit packages shall be packages in a close-fitting box conforming to PPP-B-636, class weather-resistant. Close and seal in accordance with the appendix to PPP-B-636.

5.1.2 Level B.

5.1.2.1 Unit packaging. The unit packaging shall be as specified in paragraph 5.1.1.1.

5.1.2.2 Intermediate packaging. Twelve unit packages shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-636, PPP-B-665 or PPP-B-676. The box shall be secured to prevent accidental opening.

5.1.3 Level C.

5.1.3.1 Unit packaging. Each ribbon shall be packaged in accordance with the manufacturers standard practice, providing that this insures protection for the ribbons during shipment and provides for safe delivery to its destination.

5.1.3.2 Intermediate packaging. Twelve unit packages shall be packed in accordance with the manufacturers standard practice, providing that this insures protection for the ribbons during shipment and provides for safe delivery to its destination.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified (see 6.2), forty intermediate packages (480 packaged ribbons) shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant, grade V3c or V3s. Close, seal and strap the box in accordance with the appendix to PPP-B-636.

5.2.2 Level B. The packaging shall be as specified in paragraph 5.2.1 except that the box shall conform to PPP-B-636, class-domestic, grade 275 minimum.

5.2.3 Level C. Ribbons packaged as specified in paragraph 5.1.3 shall be packed in a manner that will assure carrier acceptance and safe arrival at destination in compliance with the National Motor Freight Classification or the Uniform Freight Classification.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order and those specified in 5.3.3, all interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order and those specified in 5.3.3, all interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Special marking. In addition to the markings required by 5.3.1 or 5.3.2, each unit and intermediate package shall be marked with the color, composition, complete line of machines on which the ribbons may be used and degree of inking. Additionally, each shipping container shall also be marked with the date of manufacture (month and year) expressed numerically, followed by the letter "P"; e.g.; 12-71-P.

6. NOTES

6.1 Intended use. Ribbons covered by this specification are intended for use with computing and recording machines (see 1.1).

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Composition of ribbon required. (See 1.2.)
(c) Color of ribbon. Specify when other than equal color width of two-colored ribbons are desired. (See 3.6.)
(d) Degree of inking if other than heavy inking is desired for composition A (cotton) ribbons, and medium-heavy inking is desired for composition B (nylon) ribbons. (See 3.6.1.)
(e) Length of ribbon. (See 3.7.1.)
(f) Width of ribbon. (See 3.7.2.)
(g) Make, and model (if applicable) of machine on which ribbon is to be used. (See 3.9.)
(h) Level of packaging and packing required. (See 5.1 and 5.2.)
(i) Number of units in level A packaging and packing if other than that specified is desired. (See 5.1.1.2 and 5.2.1.)
(j) Marking requirements. (See 5.3.)

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, **and mail to: Gen. Services Admin., FSS (FMSO), Washington, D.C. 20406**

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

DDD-R-00271B (GSA-FSS) - Ribbons; Computing and Recording Machines

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

FORM GSA FEB 69 2200

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